

Coordinated Traffic Corridor Simulation

Assignment 2

Group Mehmet

Mehmet Kaan Ünsel

Ali Ayhan Günder

Mustafa Öztürkmen

Contents

1	Model Logic and Assumptions	3
1.1	Network Topology	3
1.2	Signal Control	3
1.3	Queue Discipline and Priority	3
1.4	A – B Link Model	3
1.5	Emergency Preemption Protocol	3
1.6	Warm-Up Treatment	4
1.7	Random-Number Streams	4
1.8	Conceptual Model	5
2	Experimental Design	5
3	Results	5
4	Comparative Discussion	7
4.1	Effect of Coordination (S1 vs. S2)	7
4.2	Effect of Emergency Preemption (S2 vs. S3)	7
4.3	Most Affected KPI	8
5	Verification and Validation	8
5.1	Internal Consistency Checks	8
5.2	Downstream Blocking Validation	8
5.3	Experimental-Design Verification	8
6	Conclusion	9

1 Model Logic and Assumptions

1.1 Network Topology

We modelled the corridor as a discrete-event system in SimPy with two intersections, A and B, a one-directional A – B link, and four active approaches: A N/S, A W/E, B N/S, and B W/E. Southbound entries at A follow a Poisson process with rate $\lambda = 900$ veh/h and are split into cars, buses, and emergency vehicles using the required proportions. Independent W/E arrivals at A and B each follow Poisson processes with rate $\lambda_{WE} = 400$ veh/h. Turning decisions are sampled independently at A and B according to the assignment probabilities.

1.2 Signal Control

Each intersection is controlled by a **fixed-time two-phase signal cycle**: N/S green $\rightarrow$ all-red clearance $\rightarrow$ W/E green $\rightarrow$ all-red clearance. The coordinated scenario uses the required **20 s offset** so that B's N/S green starts one free-flow travel time after A's N/S green.

1.3 Queue Discipline and Priority

Queue discipline is *explicit* rather than hidden inside a resource abstraction because service depends jointly on signal state, queue priority, and downstream link availability.

1. **Emergency vehicles** – highest priority (preemptive trigger)
2. **Buses** – second priority (non-preemptive; applied when next service starts)
3. **Regular cars** – lowest priority

1.4 A – B Link Model

The A – B link is represented as a finite-capacity point queue with $N_{\max} = 40$ and travel time

$$t_{\text{travel}}(n) = \frac{20}{1 - n/40},$$

augmented by a small multiplicative noise term sampled from a dedicated RNG stream. Non-emergency northbound vehicles at A can only enter the link when A is in N/S green and the link is not full. Emergency vehicles bypass the blocking rule exactly as required and can enter even if the link is already full; they still increase the link occupancy.

1.5 Emergency Preemption Protocol

Emergency preemption is implemented by:

1. Interrupting the current non-N/S phase,
2. Inserting a 4 s all-red clearance,
3. Forcing a 10 s N/S green, and
4. Resuming the interrupted phase remainder.

We report *recovery time* as the resumed interrupted-phase remainder after the emergency green ends. This matches the assignment definition more closely than timing from the original preemption trigger.

1.6 Warm-Up Treatment

Time-persistent KPIs (queues and link occupancy) are integrated only over $t \geq 900$ s. Through-delay observations are recorded only for southbound vehicles that entered the system after warm-up; W/E delay observations are recorded only for W/E arrivals that joined after warm-up. Throughput counts all northbound exits occurring after warm-up because it is a post-warm-up *flow-rate* KPI.

1.7 Random-Number Streams

Table 1. Random-number stream assignment. All streams are created from `numpy.random.SeedSequence([20260407, scenario_id, replication_id])`.

Stream	Purpose
0	Southbound interarrival times at A
1	W/E interarrival times at A
2	W/E interarrival times at B
3	Southbound vehicle classification
4	Turning decisions at A
5	Turning decisions at B
6	Service times at A
7	Service times at B
8	Link travel-time noise

1.8 Conceptual Model

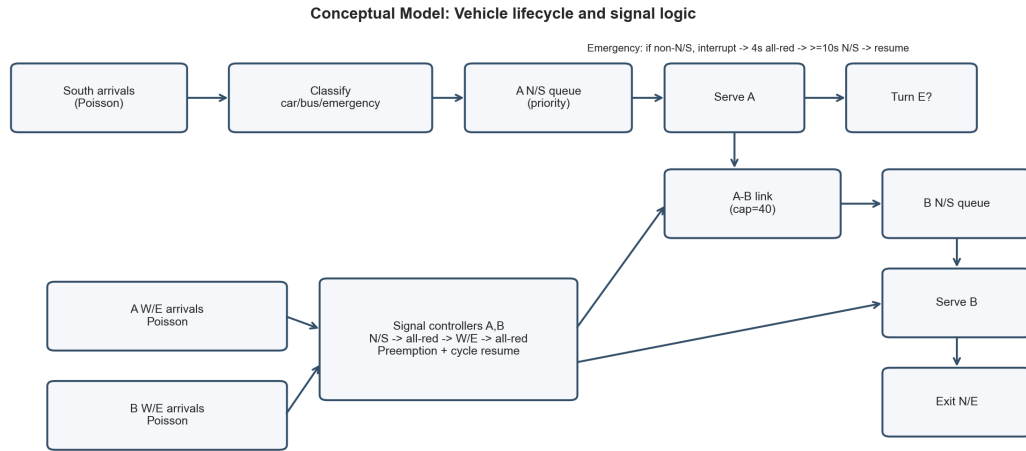


Figure 1. Conceptual model diagram generated from the simulation notebook, illustrating the two-intersection corridor with the A – B link, signal phases, and vehicle routing logic.

2 Experimental Design

We ran the three required scenarios:

Table 2. Scenario definitions.

ID	Description	Coordination	Emergencies
S1	Uncoordinated, no emergencies	Off	Off
S2	Coordinated, no emergencies	On	Off
S3	Coordinated, with emergencies	On	On

Each scenario was simulated for **7 200 s** (2 hours), the first **900 s** were discarded as warm-up, and **20 independent replications** were run. Point estimates are reported as replication means and 95 % confidence intervals are computed as $\bar{x} \pm 1.96 \times s / \sqrt{n}$.

3 Results

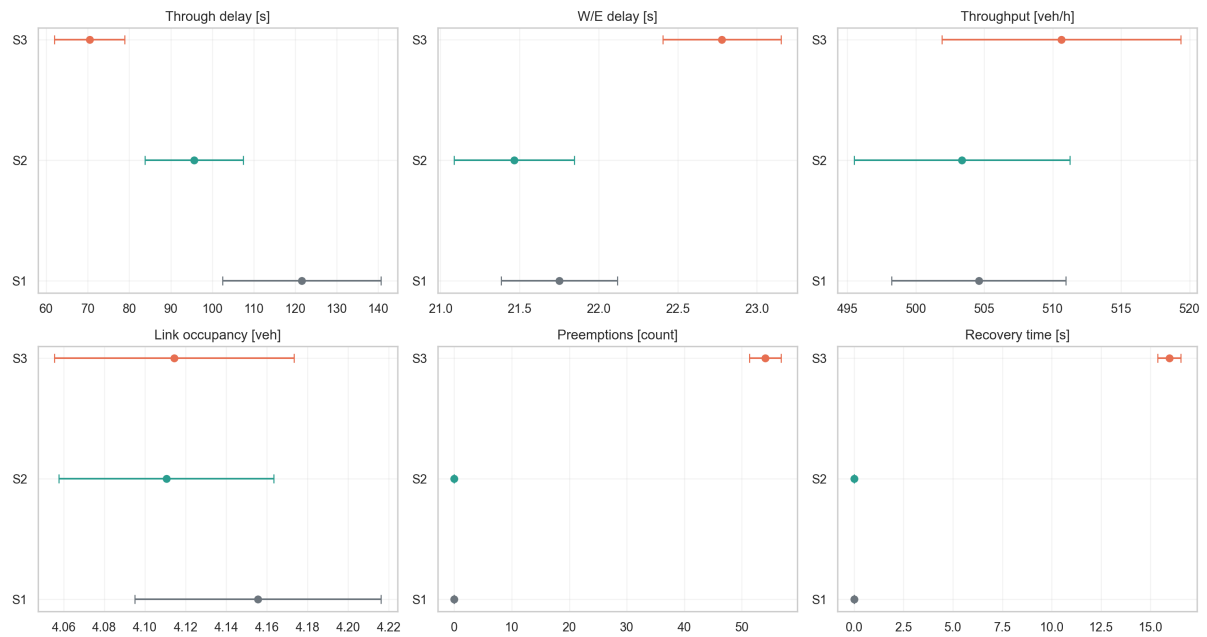
Table 3 summarises the main operational KPIs. Table 4 reports average and maximum queue lengths for each approach. Figures 2 and 3 visualise the same results using compact comparative plots saved by the notebook.

Table 3. Main KPI summary – mean [95 % CI].

KPI	S1 Uncoord.	S2 Coord.	S3 Coord. + Emerg.
Avg delay (through S→N) [s]	121.57 [102.45, 140.69]	95.58 [83.73, 107.44]	70.41 [61.95, 78.88]
Avg delay (W→E) [s]	21.75 [21.38, 22.12]	21.47 [21.09, 21.85]	22.78 [22.40, 23.15]
Throughput NB at B [veh/h]	504.60 [498.23, 510.97]	503.37 [495.48, 511.26]	510.63 [501.90, 519.35]
Avg link occupancy [veh]	4.16 [4.10, 4.22]	4.11 [4.06, 4.16]	4.11 [4.06, 4.17]
Blocking frequency [frac. cycles]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Preemptions per replication	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	54.05 [51.29, 56.81]
Recovery time [s]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	15.95 [15.36, 16.53]

Table 4. Queue-length KPI summary – mean [95 % CI].

KPI	S1 Uncoord.	S2 Coord.	S3 Coord. + Emerg.
Avg queue A N/S [veh]	24.34 [19.48, 29.20]	21.29 [18.24, 24.33]	13.25 [11.19, 15.31]
Max queue A N/S [veh]	62.10 [53.22, 70.98]	56.60 [50.05, 63.15]	41.90 [36.35, 47.45]
Avg queue B N/S [veh]	4.57 [4.39, 4.75]	2.02 [1.90, 2.14]	3.22 [2.85, 3.58]
Max queue B N/S [veh]	18.65 [17.56, 19.74]	13.30 [11.93, 14.67]	19.25 [18.13, 20.37]
Avg queue A W/E [veh]	2.40 [2.32, 2.49]	2.40 [2.31, 2.49]	2.58 [2.49, 2.68]
Max queue A W/E [veh]	12.60 [11.78, 13.42]	13.30 [12.23, 14.37]	14.10 [13.28, 14.92]
Avg queue B W/E [veh]	2.45 [2.36, 2.54]	2.39 [2.33, 2.45]	2.53 [2.44, 2.62]
Max queue B W/E [veh]	12.80 [12.12, 13.48]	12.85 [11.79, 13.91]	12.65 [12.22, 13.08]

**Figure 2.** 95 % confidence-interval comparison for the main KPIs across the three scenarios. The horizontal bars represent the CI width; dots mark replication means.

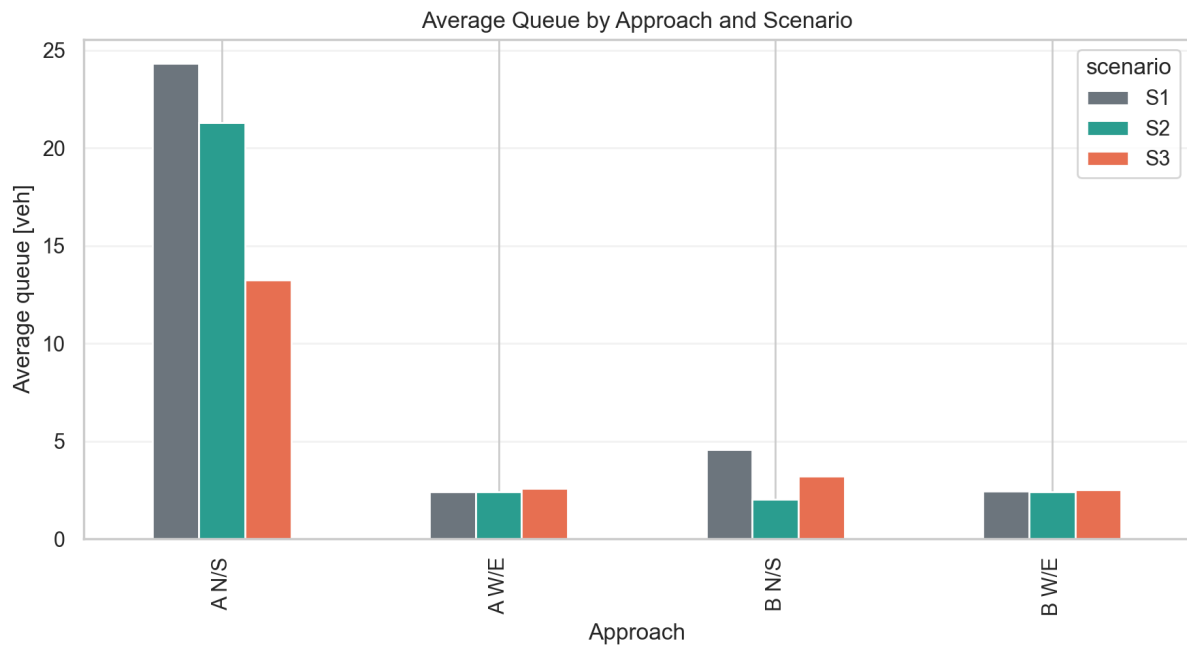


Figure 3. Queue-profile comparison across scenarios, showing both average and maximum queue lengths for every approach.

4 Comparative Discussion

4.1 Effect of Coordination (S1 vs. S2)

Coordination reduces southbound through-delay by **21.4 %** and cuts the average B N/S queue by **55.9 %**.

This is the main benefit expected from a 20 s offset: vehicles released from A arrive at B closer to the start of B's N/S green, thereby encountering shorter queues and less red-time waiting. The throughput effect is negligible (-0.24%), which suggests that A remains the practical bottleneck and coordination mainly *reduces waiting* rather than increasing discharge capacity.

4.2 Effect of Emergency Preemption (S2 vs. S3)

Relative to S2, S3 changes the performance mix rather than hurting every KPI uniformly:

- **Through-delay drops by another 26.3 %** because preemptions create bonus N/S green time that also serves non-emergency corridor vehicles waiting behind the emergency unit.
- **W/E delay rises by 6.1 %** and A W/E average queue rises by 7.7 % due to interrupted W/E phases.
- **B N/S average queue rises by 59.5 %**: preemptions release extra A platoons that reach B in bursts and bunch the downstream queue even while corridor delay improves on average.

Emergency preemption improves the priority corridor but does so by *redistributing* delay away from the emergency path and toward interrupted side-street traffic and downstream platoon build-up.

4.3 Most Affected KPI

Among the headline KPIs in Table 3, **through-delay** shows the largest percentage change between S2 and S3. Among the adverse effects specifically, the most visible degradation is the increase in cross-street delay and the sharp growth of the BN/S queue. In other words, preemption improves the priority corridor strictly at the expense of non-priority movements.

5 Verification and Validation

5.1 Internal Consistency Checks

We used several sanity checks in the notebook:

- Queue lengths and link occupancy remain *non-negative* at all times.
- Blocking frequency always lies in $[0, 1]$.
- Preemption counts are exactly zero in S1 and S2 while remaining positive in S3.

These checks verify that the core bookkeeping and scenario toggles behave as expected.

5.2 Downstream Blocking Validation

In the required scenarios the observed blocking frequency is zero for all 20 replications. This is **not a coding bug**. With the assignment parameters, A and B have symmetric N/S green times and identical single-server mean service times, so the maximum A→link discharge rate is effectively the same as the maximum B drain rate. Under that symmetry the link does not fill before A itself becomes the active bottleneck.

To validate that spillback logic exists and is correctly coded, we ran a controlled **stress test** in the notebook with B's N/S green shortened to 10 s and the link capacity reduced to 20 vehicles. Under that forced-bottleneck case the model produces positive blocking frequency and elevated link occupancy, confirming that the blocking rule is active even though KPI 5 remains zero in the official scenarios.

5.3 Experimental-Design Verification

We verified the experimental design itself:

- Replication-level 95 % confidence intervals are produced for every KPI.
- RNG streams are separated by source (Table 1).

- Warm-up treatment is applied consistently to both time-persistent and entity-based metrics.

These checks give us confidence that the reported scenario differences are due to model logic rather than bookkeeping artefacts.

6 Conclusion

This study demonstrates that a relatively simple coordination mechanism – a 20 s green-wave offset – materially improves corridor performance by reducing southbound through-delay by over 20 % and halving the downstream BN/S queue. Emergency preemption further boosts corridor delay performance, but at the cost of increased cross-street delays and downstream queue volatility. The simulation model was verified through internal consistency checks, a dedicated blocking stress test, and proper experimental-design safeguards including independent RNG streams and a rigorous warm-up protocol. Together, these results highlight that signal coordination is an effective but inherently trade-off-laden strategy: corridor efficiency gains come at the expense of non-priority movements, and emergency preemption amplifies this redistribution effect.